PS5 Report

2. Test the method on simple hard-coded graphs and input strings (e.g., from programming drill, along with others you make up). In your report, discuss the tests and how they convinced you of your code's correctness:

- To test our code, we hard-coded the graph from recitation 7. However, to make it work, we slightly altered the values attached to the edges and vertices by following the scoring conventions in this assignment. To create the graph, we inserted each of the vertices from the graph into transitions, along with a map of their adjacent vertices and the weight of the edge that connects to that vertex. Next, we inserted each of the vertices from the graph into observations, along with a map of all their words and the values of those words. We used this graph to test our code, with sentences like “chase and get and dog and cat” or “dog and cat watch and chase”. These tests confirmed that our viterbi algorithm was working correctly, as the output was correct.

6. In a short report, provide some new example sentences that are tagged as expected and some that aren't, discussing why. Also discuss your overall testing performance, and how it depends on the unseen-word penalty (and any other parameters you use).

- Example sentences:
    - Ashok and Mithun completed Problem Set 5 .
        - Result: Ashok/ADV and/CNJ Mithun/V completed/VN Problem/N Set/V 5/NUM ./.
        - This test did not tag as expected, because the training data does not contain our names, which forces the algorithm to assign the unseen word penalty. This could lead to incorrect tag assignments since the algorithm may have prioritized tags with higher prior probabilities or better transitions rather than the expected parts of speech.
    - hi , how are you ?
        - Result: hi/UH ,/, how/WH are/V you/PRO ?/.
        - This test worked as expected, likely because these are all very common words, which increases the likelihood of the algorithm correctly assigning them to their tags
    - we are at Dartmouth college .
        - Result: we/PRO are/V at/P Dartmouth/DET college/N ./.
        - This test did not work as expected, as it thought Dartmouth was a determiner, not a proper noun. Similar to the first test, this happened because Dartmouth was not in the training data, so it made its best guess, which turned out to be incorrect
- Overall testing performance:
    - Overall, our testing performance was quite good, with our simple test producing an accuracy level of 86.49, and our brown test producing an accuracy level of 96.47 (unseen word penalty of -100). The unseen word penalty has a large impact on the results. When the penalty is -10, our brown accuracy level was only 88.99%. Without a penalty, our simple accuracy level was just 24.32% and

brown was even worse at 6.21%. When we changed the penalty to -1000, we got the exact same results as when the penalty was -100, so increasing the penalty too heavily isn't helpful either.